

DIMENSION-DEPENDENT BOUNDS FOR THE SDIEP VIA PHASE OPTIMISATION AND PALEY-TYPE CONSTRUCTIONS

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*In memoriam: Miroslav Fiedler (1926–2015),
on the occasion of the tenth anniversary of his passing*

ABSTRACT. We develop structured numerical linear algebra algorithms for constructing symmetric doubly stochastic matrices with prescribed Suleĭmanova spectrum. The algorithms use the Schur realisation $P(\Lambda) = Q\Lambda Q^\top$, where Q is a structured orthogonal basis with Perron column $\mathbf{1}/\sqrt{n}$; entrywise non-negativity is certified by a coherence constant that is cheap to compute and sharp for the bases considered here. For the canonical cycle basis we compute the exact size-dependent threshold

$$\delta_n = 1 - \frac{1}{2 \cos^2\left(\frac{\pi}{4n}\rho(n)\right)}, \quad \rho(n) \in \{0, 1, 2, 4\} \text{ determined by } n \bmod 8,$$

which improves the previously used $1/2$ bound if and only if $8 \nmid n$, and we prove sharpness for that template. We then introduce an optimally phase-aligned cycle basis for which

$$\delta_n^{(\text{ph})} = \begin{cases} 1 - \frac{1}{2 \cos^2(\pi/n)}, & n \equiv 0 \pmod{4}, \\ 1 - \frac{1}{2 \cos^2(\pi/2n)}, & n \equiv 2 \pmod{4}, \\ 1 - \frac{1}{2 \cos^2(\pi/4n)}, & n \text{ odd}, \end{cases}$$

so $\delta_n^{(\text{ph})} < 1/2$ for every $n \geq 3$. In Hadamard orders, including Sylvester–Walsh and Paley families, the coherence is optimal, $M(Q) = 1$, and every strict Suleĭmanova list is realised. The numerical part of the paper gives reproducible tests of the certificates, floating-point residuals, computational cost, Monte-Carlo success probabilities, and two application-oriented experiments: design of reversible Markov chains with prescribed mixing rates and average-consensus iterations with certified non-negative weights.

1. INTRODUCTION

A real square matrix is called doubly stochastic if it has non-negative entries and all of its row and column sums are equal to one. The Symmetric Doubly Stochastic Inverse Eigenvalue Problem (SDIEP) asks for the necessary and sufficient conditions under which a given list of real numbers

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$\sigma = (\lambda_1, \lambda_2, \dots, \lambda_n)$ can be the spectrum of an $n \times n$ symmetric doubly stochastic matrix [1]. By the Perron–Frobenius theorem, any such matrix must have a Perron eigenvalue of 1, corresponding to the eigenvector of all ones. This fixes $\lambda_1 = 1$ and, by Gershgorin’s circle theorem, bounds all other eigenvalues to the interval $[-1, 1]$.

A significant body of research focuses on the realisability of a *Suleĭmanova spectrum*, a list of the form $(1, \lambda_2, \dots, \lambda_n)$ where each $\lambda_j \leq 0$ for $j \geq 2$ and the trace is non-negative, $1 + \sum_{j=2}^n \lambda_j \geq 0$. A powerful constructive method for tackling this problem is the Schur template: given a diagonal matrix $\Lambda = \text{diag}(1, \lambda_2, \dots, \lambda_n)$, one constructs a candidate matrix $P(\Lambda) = Q\Lambda Q^\top$. If Q is an orthogonal matrix whose first column is the Perron eigenvector $q_0 = \mathbf{1}/\sqrt{n}$, the resulting matrix $P(\Lambda)$ is automatically symmetric, doubly stochastic, and has the desired spectrum.

Algorithmic template (Schur realisation).

Given a basis Q with $q_0 = \mathbf{1}/\sqrt{n}$ and a list $\Lambda = \text{diag}(1, \lambda_2, \dots, \lambda_n)$:

- (1) Form $Y := Q\Lambda$ (cost $O(n^2)$)
- (2) Form $P := YQ^\top$ (dense GEMM, cost $O(n^3)$)
- (3) (Optional) Check entrywise nonnegativity: $\min_{k,\ell} P_{k\ell} \geq -\tau$ with tolerance τ .

For a chosen Q , Lemma 10 gives the sum-only sufficient condition $1 + \sum_{j=2}^n \lambda_j > 1 - \frac{1}{M(Q)}$ that guarantees $P \geq 0$. Lower $M(Q)$ *tightens* this certificate.

The entire problem is thus reduced to a single, challenging question: for a given spectrum Λ and a chosen basis Q , what conditions guarantee that $P(\Lambda)$ is entrywise non-negative?

The papers [12] and [13] seem to be foundational for the problem. Recent work has approached this question from several angles. One approach provides general sufficient conditions that depend only on the sum of the non-Perron eigenvalues. A key result in this vein was provided by Gnacik and the author [1], who used the eigenbasis of the n -cycle graph for Q to show that any Suleĭmanova spectrum is realisable if $1 + \sum_{j=2}^n \lambda_j > 1/2$. It was noted, however, that this bound appeared to be sharp for this basis, particularly in dimensions where n is a multiple of 8. A different approach, seen in the work of Sarkhoni *et al.* [14], forgoes general sum-based conditions and instead derives necessary and sufficient conditions for highly structured spectra, such as $(1, 0, \dots, 0, \lambda_n)$, by constructing a bespoke orthogonal matrix Q tailored to that specific list.

In this paper, we unify and advance these perspectives. We begin by revisiting the cycle-walk basis from [1], replacing their baseline bound with a precise, dimension-dependent threshold that confirms the ‘ $8 \mid n$ ’ barrier is an artefact of the basis, not the problem itself. We then resolve this issue by introducing an optimally phase-aligned cycle basis that yields an improved bound, strictly better than $1/2$ for all dimensions $n \geq 3$. Finally, we generalise the underlying principle to a criterion based on basis *coherence*, showing that Walsh–Hadamard bases from elementary abelian 2-groups, and more generally Hadamard orders such as Paley families, achieve the ideal coherence. After the theoretical development we give numerical tests, residual checks, timing measurements, and application experiments for Markov-chain and average-consensus design.

Let $\sigma = (1, \lambda_2, \dots, \lambda_n)$ with $\lambda_j \in [-1, 0]$ for $j \geq 2$. Following Paparella [10], we call σ a *Suleřmanova spectrum* and say it is δ -*Suleřmanova* if

$$1 + \sum_{j=2}^n \lambda_j > \delta.$$

In [1] a Markov-chain-inspired template was introduced: take $Q = [w_0, \dots, w_{n-1}]$ to be a real orthonormal eigenbasis of the simple random walk on the n -cycle, so $w_0 = \mathbf{1}/\sqrt{n}$ and, for $k \geq 1$,

$$w_k(j) = \sqrt{\frac{2}{n}} \sin\left(\frac{2\pi k j}{n} + \frac{\pi}{4}\right) \quad (j = 0, \dots, n-1).$$

Setting $P(\Lambda) = Q \operatorname{diag}(1, \lambda_1, \dots, \lambda_{n-1}) Q^\top$ gives a symmetric matrix with unit row sums. Using $S_j(k) := \sin\left(\frac{2\pi k j}{n} + \frac{\pi}{4}\right)$ one has

$$p_{kl} = \frac{1}{n} \left(1 + 2 \sum_{j=1}^{n-1} \lambda_j S_j(k) S_j(l) \right).$$

The crude estimate $S_j(k)S_j(l) \leq 1$ and $\lambda_j \leq 0$ yields the sufficient condition $1 + \sum_{j=2}^n \lambda_j > \frac{1}{2}$. It was also observed that when $8 \mid n$ there are j, k with $S_j(k) = 1$, so this estimate cannot beat $\frac{1}{2}$ for those sizes, and exhibited a 5×5 spectrum at sum 0.48 violating entrywise positivity. This suggests trying other orthogonal choices of Q to reduce the worst interaction $n |q_j(k)q_j(l)|$ (all from [1]).

In the present paper, we keep the Schur template $P(\Lambda) = Q\Lambda Q^\top$ and:

- compute *exactly* the worst coefficient for the canonical cycle basis, obtaining a closed-form δ_n that improves $\frac{1}{2}$ iff $8 \nmid n$, with a matching sharpness example;
- introduce an *optimally phase-aligned* cycle basis (still orthogonal, same Perron column) giving the *correct* piecewise bound $\delta_n^{(\text{ph})} < \frac{1}{2}$ for every $n \geq 3$, which strictly improves the $8 \mid n$ case and is optimal within the cycle family;
- prove a general ‘coherence $\Rightarrow$ sufficient δ ’ lemma and apply it to Walsh–Hadamard bases from elementary abelian 2-groups and to Hadamard orders (via Paley), obtaining $\delta = 0$ *in those orders only*. Here ‘ $\delta = 0$ ’ means the *strict* condition $1 + \sum \lambda_j > 0$ suffices; this does not contradict obstructions on the boundary $1 + \sum \lambda_j = 0$ (e.g., $(1, 0, \dots, 0, -1)$ in odd dimensions);
- give a numerical linear algebra implementation path: dense Schur construction, coherence-based certification, residual checks, timing experiments, Monte-Carlo stress tests, and two application tests in reversible Markov chains and average consensus.

2. STRUCTURED SCHUR REALISATION AND NUMERICAL CERTIFICATION

The constructive part of the SDIEP can be viewed as an inverse eigensolver with a non-negativity certificate. Instead of starting from a matrix and computing its eigenvalues, we start from the

desired eigenvalues and choose a structured orthogonal basis in which the Perron vector is fixed. The numerical task is then to construct

$$P = Q\Lambda Q^\top, \quad Q^\top Q = I, \quad q_0 = \frac{1}{\sqrt{n}},$$

and to certify that the entries of P are non-negative. The following algorithm is the implementation used in the experiments of Section 8.

Algorithm 1 (Structured Schur realisation with coherence certificate). Given a list $\sigma = (1, \lambda_2, \dots, \lambda_n)$ with $\lambda_j \leq 0$, a structured basis family, and a numerical tolerance τ :

(1) Construct an orthogonal matrix $Q = [q_0, \dots, q_{n-1}]$ with $q_0 = \mathbf{1}/\sqrt{n}$.

(2) Compute, or evaluate from a closed formula, the coherence

$$M(Q) = n \max_{1 \leq j \leq n-1} \|q_j\|_\infty^2.$$

(3) Let $T = 1 + \sum_{j=2}^n \lambda_j$. If $T > 1 - 1/M(Q)$, record the analytic certificate

$$\min_{k,\ell} P_{k\ell} \geq \frac{1}{n} (1 + M(Q)(T - 1)) > 0.$$

(4) Form $Y = Q\Lambda$ by diagonal scaling and then $P = YQ^\top$ by a dense matrix–matrix product.

(5) Report the diagnostic quantities

$$\|P\mathbf{1} - \mathbf{1}\|_\infty, \quad \|P - P^\top\|_F, \quad \min_{k,\ell} P_{k\ell}, \quad \max_i |\mu_i(P) - \lambda_i|,$$

where the last residual compares sorted spectra. Accept the numerical realisation if the analytic certificate holds or, in exploratory tests, if $\min_{k,\ell} P_{k\ell} \geq -\tau$.

For an explicitly stored dense basis, the diagonal scaling costs $O(n^2)$ operations and the product $YQ^\top$ costs $O(n^3)$ operations. These costs are appropriate when the full transition or averaging matrix is needed. If only products Px are required, the same factorisation gives

$$Px = Q(\Lambda(Q^\top x)),$$

so the cost is determined by fast applications of Q and $Q^\top$: Walsh–Hadamard bases allow fast Walsh–Hadamard transforms, and the cycle bases can be applied through Fourier-type transforms after passing through the corresponding real representation. Thus the constructions below are not merely existence proofs; they provide structured matrix generators and matrix-free linear operators with verifiable stochasticity and spectral residuals.

3. THE CYCLE-WALK BASIS

From now on we reindex the list as $\sigma = (\lambda_0, \lambda_1, \dots, \lambda_{n-1})$ with $\lambda_0 = 1$ and $\lambda_j \in [-1, 0]$ for $1 \leq j \leq n-1$, and use the cycle eigenbasis $Q = [w_0, \dots, w_{n-1}]$ described above.

Lemma 2. *Let Q and Λ be as above. Then*

$$(1) \quad p_{kl} = \frac{1}{n} \left(1 + 2 \sum_{j=1}^{n-1} \lambda_j S_j(k) S_j(l) \right), \quad S_j(k) := \sin\left(\frac{2\pi k j}{n} + \frac{\pi}{4}\right).$$

If $\lambda_j \leq 0$, writing

$$c_n := \max_{1 \leq j \leq n-1} \max_{0 \leq k, l \leq n-1} S_j(k) S_j(l),$$

we have the bound

$$(2) \quad p_{kl} \geq \frac{1}{n} \left(1 + 2c_n \sum_{j=1}^{n-1} \lambda_j \right) \quad (0 \leq k, l \leq n-1).$$

Proof. Orthogonality and $w_0 = \mathbf{1}/\sqrt{n}$ give $P(\Lambda)\mathbf{1} = \mathbf{1}$. Expanding $Q\Lambda Q^\top$ yields (1). Because $\lambda_j \leq 0$ and $S_j(k)S_j(l) \leq c_n$ for each j , multiplying the latter inequality by λ_j reverses it, hence $\sum_{j=1}^{n-1} \lambda_j S_j(k)S_j(l) \geq c_n \sum_{j=1}^{n-1} \lambda_j$, and (2) follows. $\square$

Lemma 3 (Exact value of c_n and the mod-8 mechanism). *Define*

$$\Delta_n := \min_{m, t \in \mathbb{Z}} \left| \frac{2\pi m}{n} - \frac{(2t+1)\pi}{4} \right|.$$

Then

$$c_n = \cos^2 \Delta_n \quad \text{and} \quad \Delta_n = \frac{\pi}{4n} \rho(n), \quad \rho(n) = \begin{cases} 0, & n \equiv 0 \pmod{8}, \\ 1, & n \equiv 1, 3, 5, 7 \pmod{8}, \\ 2, & n \equiv 2, 6 \pmod{8}, \\ 4, & n \equiv 4 \pmod{8}. \end{cases}$$

Moreover, the maximum in c_n is attained with $k = l$.

Proof. For fixed j with $\gcd(j, n) = 1$, the multiset $\{2\pi k j/n + \pi/4 : 0 \leq k \leq n-1\}$ is a permutation of the grid $\{2\pi m/n + \pi/4 : 0 \leq m \leq n-1\}$. Hence

$$\max_k |S_j(k)| = \max_m \left| \sin \left(\frac{2\pi m}{n} + \frac{\pi}{4} \right) \right| = \sin \left(\frac{\pi}{2} - \Delta_n \right) = \cos \Delta_n,$$

where Δ_n is the minimal angular distance from this grid to the odd half-angles. Therefore $c_n = (\max_{j,k} |S_j(k)|)^2 = \cos^2 \Delta_n$, and the maximum occurs already for $k = l$ at a peak. As

$$\frac{2\pi m}{n} - \frac{(2t+1)\pi}{4} = \frac{\pi}{4n} (8m - (2t+1)n)$$

implies $\Delta_n = \frac{\pi}{4n} \min_{m, t} |8m - (2t+1)n|$. We now analyse residues of n modulo 8.

- $n \equiv 0 \pmod{8}$. Then $(2t+1)n$ is divisible by 8 for every t , so choosing $m = (2t+1)n/8$ gives the minimum 0. Thus $\rho(n) = 0$.
- n odd. Here $\gcd(n, 8) = 1$. By Bézout's theorem, there exist integers m, t with $8m - (2t+1)n = 1$. Hence, the minimum positive value is 1 and $\rho(n) = 1$.
- $n \equiv 2, 6 \pmod{8}$. Write $n = 2k$ with k odd. Then

$$8m - (2t+1)n = 2(4m - (2t+1)k)$$

is always even. Moreover $4m - (2t+1)k$ is always odd, since $4m$ is even and $(2t+1)k$ is odd, so $8m - (2t+1)n$ is never divisible by 4. Hence the minimum positive value is

at least 2. Conversely, since $\gcd(4, k) = 1$, there exist integers m, r with $4m - rk = 1$. Reducing modulo 2 shows that r is odd, say $r = 2t + 1$, and then

$$8m - (2t + 1)n = 2.$$

Thus $\rho(n) = 2$.

- $n \equiv 4 \pmod{8}$. Write $n = 4k$ with k odd. Then

$$8m - (2t + 1)n = 4(2m - (2t + 1)k)$$

is always divisible by 4. Since $2m - (2t + 1)k$ is odd, it is never divisible by 2, so $8m - (2t + 1)n$ is never divisible by 8. Hence the minimum positive value is at least 4. Conversely, since $\gcd(2, k) = 1$, there exist integers m, r with $2m - rk = 1$. Again r must be odd, say $r = 2t + 1$, and then

$$8m - (2t + 1)n = 4.$$

Thus $\rho(n) = 4$.

□

Theorem 4 (Dimension-dependent bound for the cycle basis). *With Δ_n as above, set*

$$\delta_n := 1 - \frac{1}{2 \cos^2 \Delta_n} = \frac{1}{2} - \frac{1}{2} \tan^2 \Delta_n.$$

If $1 + \sum_{j=2}^n \lambda_j > \delta_n$, then $P(\Lambda) = Q\Lambda Q^\top$ is symmetric and doubly stochastic. In particular $\delta_n < \frac{1}{2}$ if and only if $8 \nmid n$, whereas $\delta_n = \frac{1}{2}$ when $8 \mid n$.

Proof. By Lemma 2, for every k, l ,

$$p_{kl} \geq \frac{1}{n} \left(1 + 2c_n \sum_{j=1}^{n-1} \lambda_j \right), \quad c_n = \cos^2 \Delta_n.$$

by Lemma 3. Hence $p_{kl} \geq 0$ whenever $\sum_{j=1}^{n-1} \lambda_j > -1/(2c_n)$, i.e., when

$$1 + \sum_{j=2}^n \lambda_j > 1 - \frac{1}{2 \cos^2 \Delta_n} = \delta_n,$$

which establishes entrywise nonnegativity under the stated hypothesis.

Since Q is orthogonal and Λ is (real and) diagonal (hence $\Lambda^\top = \Lambda$),

$$P(\Lambda)^\top = (Q\Lambda Q^\top)^\top = (Q^\top)^\top \Lambda^\top Q^\top = Q\Lambda Q^\top = P(\Lambda),$$

so $P(\Lambda)$ is symmetric.

Let $q_0, \dots, q_{n-1}$ be the columns of Q and let $e_0 = (1, 0, \dots, 0)^\top$. Since $q_0 = \mathbf{1}/\sqrt{n}$ and $q_j \perp q_0$ for $j \geq 1$,

$$Q^\top \mathbf{1} = (\langle q_0, \mathbf{1} \rangle, \dots, \langle q_{n-1}, \mathbf{1} \rangle)^\top = (\sqrt{n}, 0, \dots, 0)^\top = \sqrt{n} e_0.$$

Therefore

$$P(\Lambda) \mathbf{1} = Q\Lambda Q^\top \mathbf{1} = Q\Lambda(\sqrt{n} e_0) = \sqrt{n} Q(\Lambda e_0) = \sqrt{n} Q e_0 = \sqrt{n} q_0 = \sqrt{n} \frac{\mathbf{1}}{\sqrt{n}} = \mathbf{1}.$$

Thus every row sum equals 1. Finally,

$$\begin{aligned} \mathbf{1}^\top P(\Lambda) &= \mathbf{1}^\top Q \Lambda Q^\top = (Q^\top \mathbf{1})^\top \Lambda Q^\top = (\sqrt{n} e_0)^\top \Lambda Q^\top \\ &= \sqrt{n} e_0^\top \Lambda Q^\top = \sqrt{n} e_0^\top Q^\top = (\sqrt{n} Q e_0)^\top = \mathbf{1}^\top, \end{aligned}$$

where we used that $\Lambda e_0 = e_0$ because the first diagonal entry of Λ is $\lambda_0 = 1$. Hence, every column sum also equals 1.

Combining the three parts shows that $P(\Lambda)$ is symmetric and doubly stochastic, whenever $1 + \sum_{j=2}^n \lambda_j > \delta_n$. The stated equivalence for $\delta_n < \frac{1}{2}$ versus $8 \nmid n$ follows from Lemma 3, which gives $c_n = \cos^2 \Delta_n = 1$ iff $8 \mid n$. $\square$

Proposition 5 (Sharpness for the cycle basis). *Fix n and choose indices realising*

$$c_n = \max_{j,k,l} S_j(k) S_j(l).$$

If $\lambda_{j^} = t < 0$ and $\lambda_j = 0$ for $j \neq j^*$, then*

$$p_{k^*k^*} = \frac{1}{n} (1 + 2t \cos^2 \Delta_n),$$

which is non-negative if and only if $t \geq -1/(2 \cos^2 \Delta_n)$, i.e. $1 + \sum_{j=2}^n \lambda_j \geq \delta_n$. Thus Theorem 4 is optimal for this Q among conditions depending only on $\sum_{j \geq 2} \lambda_j$.

Remark 6. We see that divisibility by 8 is the exact barrier. Indeed, by Lemma 3, the grid $\{2\pi m/n + \pi/4\}$ hits an odd half-angle iff $8 \mid n$, and then $c_n = 1$ so the factor $\frac{1}{2}$ is unavoidable for this Q . If $8 \nmid n$, the nearest grid point sits at angular distance $\Delta_n = \pi \rho(n)/(4n)$ from $\pi/2$, giving $c_n = \cos^2 \Delta_n < 1$ and hence $\delta_n < 1/2$.

4. PHASE-OPTIMISED CYCLE BASES

The eigenbasis of the n -cycle graph is not unique. For each $j \in \{1, \dots, \lfloor (n-1)/2 \rfloor\}$, the corresponding eigenspace E_j is two-dimensional, spanned by a sine and cosine vector. The canonical basis analysed in Section 3 represents just one choice of basis within these spaces, a choice that, for dimensions where $8 \mid n$, leads to an unavoidable coherence peak and a bound of $\delta_n = 1/2$. This section demonstrates that by rotating the basis in each eigenspace via an optimally chosen phase angle ϕ , we can construct a new orthogonal matrix $Q^{(\text{ph})}$. This ‘phase-optimised’ basis is designed to minimise the maximum magnitude of its entries, effectively smoothing the coherence peaks and removing the “ $8 \mid n$ ” artefact entirely. More specifically, for each $j \in \{1, \dots, \lfloor (n-1)/2 \rfloor\}$ the cycle eigenspace is two-dimensional, $E_j = \text{span}\{u_j, v_j\}$ with $u_j(k) = \sqrt{\frac{2}{n}} \cos(\frac{2\pi jk}{n})$, $v_j(k) = \sqrt{\frac{2}{n}} \sin(\frac{2\pi jk}{n})$. Any orthonormal pair in E_j is of the form

$$w_j^{(\phi)}(k) = \sqrt{\frac{2}{n}} \sin\left(\frac{2\pi jk}{n} + \phi\right), \quad \tilde{w}_j^{(\phi)}(k) = \sqrt{\frac{2}{n}} \cos\left(\frac{2\pi jk}{n} + \phi\right),$$

for some phase ϕ . When n is even and $j = n/2$, the eigenspace is one-dimensional; choosing the phase so that $|\sin(\phi)| = 1/\sqrt{2}$ yields a unit vector with $|w_{n/2}(k)| \equiv 1/\sqrt{n}$, hence $n |q_{n/2}(k) q_{n/2}(l)| \equiv 1$.

We must control *both* columns in each E_j simultaneously. Write $n' := n/\gcd(j, n)$; then $\{\frac{2\pi jk}{n}\}$ is a translate of the grid with step $2\pi/n'$.

Lemma 7 (Two-phase covering radius on a grid). *Let $n' := n/\gcd(j, n)$ and set $S := 2\pi/n'$ and $L := \pi/2$. All angles are considered reduced modulo L . Define*

$$\Delta^*(n') := \begin{cases} \pi/n', & n' \equiv 0 \pmod{4}, \\ \pi/(2n'), & n' \equiv 2 \pmod{4}, \\ \pi/(4n'), & n' \text{ odd}. \end{cases}$$

For $w_j^{(\phi)}(k) = \sqrt{\frac{2}{n}} \sin(\frac{2\pi jk}{n} + \phi)$ and $\tilde{w}_j^{(\phi)}(k) = \sqrt{\frac{2}{n}} \cos(\frac{2\pi jk}{n} + \phi)$ one has

$$\min_{\phi} \max \left\{ \|w_j^{(\phi)}\|_{\infty}, \|\tilde{w}_j^{(\phi)}\|_{\infty} \right\} = \sqrt{\frac{2}{n}} \cos(\Delta^*(n')),$$

and consequently

$$\min_{\phi} \sup_{k,l} n \max \{ |w_j^{(\phi)}(k)w_j^{(\phi)}(l)|, |\tilde{w}_j^{(\phi)}(k)\tilde{w}_j^{(\phi)}(l)| \} = 2 \cos^2(\Delta^*(n')).$$

Proof. Let $A := \{mL : m \in \mathbb{Z}\}$ with $L = \pi/2$. For any $x \in \mathbb{R}$,

$$(3) \quad \max\{|\sin x|, |\cos x|\} = \cos(\text{dist}(x, A)).$$

Indeed, reduce x modulo L to $y \in [0, L]$. Then

$$|\cos x| = |\cos y|, \quad |\sin x| = |\sin y| = |\cos(L - y)|.$$

Hence $\max\{|\sin x|, |\cos x|\} = \max\{|\cos y|, |\cos(L - y)|\} = \cos(\min\{y, L - y\})$. However $\min\{y, L - y\} = \text{dist}(x, A)$ by construction, giving (3).

Fix j and write $n' := n/\gcd(j, n)$. The set of phases

$$\Theta_j := \left\{ \frac{2\pi jk}{n} : k = 0, 1, \dots, n-1 \right\}$$

is a union of $\gcd(j, n)$ translates of the uniform grid $\{2\pi m/n' : m = 0, \dots, n' - 1\}$. Since a translation does not affect sup norms, it suffices to study the latter uniform grid. After adding a free phase ϕ and reducing modulo L we obtain

$$X_{\phi} := \left\{ \frac{2\pi m}{n'} + \phi \pmod{L} : m = 0, \dots, n' - 1 \right\} \subset \mathbb{R}/L\mathbb{Z}.$$

This set is a coset (translate) of the cyclic subgroup $G := \langle \bar{S} \rangle$ of the circle $\mathbb{R}/L\mathbb{Z}$, generated by the step $\bar{S}$ which is the class of $S := 2\pi/n'$ modulo L .

The order $r := |G|$ is the least positive integer such that $r\bar{S} = \bar{0}$, i.e.

$$rS \in L\mathbb{Z} \iff \frac{2\pi r}{n'} = \frac{\pi}{2} q \text{ for some } q \in \mathbb{Z} \iff \frac{4r}{n'} = q \in \mathbb{Z}.$$

Thus r is the least positive integer with $n' \mid 4r$, which is

$$r = \frac{n'}{\gcd(n', 4)}.$$

Consequently, the r points of G are equally spaced on the circle $\mathbb{R}/L\mathbb{Z}$ with spacing

$$(4) \quad s_{\text{eff}} = \frac{L}{r} = \frac{\pi}{2} \cdot \frac{\gcd(n', 4)}{n'} = \begin{cases} 2\pi/n', & n' \equiv 0 \pmod{4}, \\ \pi/n', & n' \equiv 2 \pmod{4}, \\ \pi/(2n'), & n' \text{ odd.} \end{cases}$$

For a fixed translate $X_\phi = \bar{\phi} + G$, define the nearest-point distance

$$d(\phi) := \min_{x \in X_\phi} \text{dist}(x, \bar{0}).$$

Since X_ϕ consists of r equally spaced points with gap $s_{\text{eff}} = L/r$, the circle $\mathbb{R}/L\mathbb{Z}$ is partitioned into r arcs of length s_{eff} whose endpoints are the points of X_ϕ . The quantity $d(\phi)$ is the distance from $\bar{0}$ to the nearest endpoint of the arc containing $\bar{0}$, so

$$d(\phi) \leq \frac{s_{\text{eff}}}{2},$$

with equality exactly when $\bar{0}$ is the midpoint of such an arc. Therefore

$$\max_{\phi} d(\phi) = \frac{s_{\text{eff}}}{2}.$$

For instance, if $x_0 \in G$ is arbitrary, choosing ϕ so that $\bar{\phi} + x_0 = \overline{s_{\text{eff}}/2}$ attains this maximum.

For any ϕ we have, by (3),

$$\max \left\{ \|w_j^{(\phi)}\|_\infty, \|\tilde{w}_j^{(\phi)}\|_\infty \right\} = \sqrt{\frac{2}{n}} \max_{x \in X_\phi} \max \{ |\sin x|, |\cos x| \} = \sqrt{\frac{2}{n}} \max_{x \in X_\phi} \cos(\text{dist}(x, \bar{0})).$$

Since $\cos$ is decreasing on $[0, \pi/2]$, the inner maximum equals $\cos(d(\phi))$. Hence

$$\min_{\phi} \max \left\{ \|w_j^{(\phi)}\|_\infty, \|\tilde{w}_j^{(\phi)}\|_\infty \right\} = \sqrt{\frac{2}{n}} \cos\left(\max_{\phi} d(\phi)\right) = \sqrt{\frac{2}{n}} \cos\left(\frac{s_{\text{eff}}}{2}\right).$$

Combining this with (4) gives the stated value $\sqrt{2/n} \cos(\Delta^*(n'))$.

Finally, for either column in the pair, we obtain

$$\min_{\phi} \sup_{k,l} n |q(k)q(l)| \leq \min_{\phi} n \|q\|_\infty^2 \leq n \cdot \left(\sqrt{\frac{2}{n}} \cos(\Delta^*(n')) \right)^2 = 2 \cos^2(\Delta^*(n')).$$

This upper bound is tight because, at a minimising shift, there is some k with

$$\text{dist}\left(\frac{2\pi j k}{n} + \phi, A\right) = d(\phi) = \frac{s_{\text{eff}}}{2},$$

hence for one of the two vectors we have

$$|q(k)| = \sqrt{\frac{2}{n}} \cos(\Delta^*(n')).$$

Taking $l = k$ shows that the value $2 \cos^2(\Delta^*(n'))$ is attained. □

Because $\Delta^*(n')$ decreases with n' , the worst case is $\gcd(j, n) = 1$, i.e., $n' = n$.

Theorem 8 (Phase-optimised coherence). *There exists an orthogonal matrix $Q^{(\text{ph})}$ with first column $\mathbf{1}/\sqrt{n}$ such that*

$$M(Q^{(\text{ph})}) = \begin{cases} 2 \cos^2(\frac{\pi}{n}), & n \equiv 0 \pmod{4}, \\ 2 \cos^2(\frac{\pi}{2n}), & n \equiv 2 \pmod{4}, \\ 2 \cos^2(\frac{\pi}{4n}), & n \text{ odd}, \end{cases}$$

and this value is optimal within the cycle family. Consequently,

$$\delta_n^{(\text{ph})} := 1 - \frac{1}{M(Q^{(\text{ph})})} = \begin{cases} 1 - \frac{1}{2 \cos^2(\pi/n)}, & n \equiv 0 \pmod{4}, \\ 1 - \frac{1}{2 \cos^2(\pi/2n)}, & n \equiv 2 \pmod{4}, \\ 1 - \frac{1}{2 \cos^2(\pi/4n)}, & n \text{ odd}, \end{cases}$$

and $0 \leq \delta_n^{(\text{ph})} < \frac{1}{2}$ for every $n \geq 3$. Moreover, $\delta_n^{(\text{ph})} = \delta_n$ unless $8 \mid n$, and in all cases $\delta_n^{(\text{ph})} \nearrow \frac{1}{2}$ as $n \rightarrow \infty$.

Proof. To show the upper bound, for each $j \in \{1, \dots, \lfloor (n-1)/2 \rfloor\}$ let us choose a phase ϕ_j that attains the minimum in Lemma 7 for $n' = n/\gcd(j, n)$, and replace the canonical pair $\{u_j, v_j\}$ by $\{w_j^{(\phi_j)}, \tilde{w}_j^{(\phi_j)}\}$. This rotation within E_j preserves orthonormality and keeps the Perron column $w_0 = \mathbf{1}/\sqrt{n}$ unchanged; doing this independently for each j yields an orthogonal matrix $Q^{(\text{ph})}$.

By Lemma 7, for each such j ,

$$\max \{ \|w_j^{(\phi_j)}\|_\infty, \|\tilde{w}_j^{(\phi_j)}\|_\infty \} = \sqrt{\frac{2}{n}} \cos(\Delta^*(n')),$$

hence for either column in this pair we have $n \|q\|_\infty^2 \leq 2 \cos^2(\Delta^*(n'))$. If n is even and $j = n/2$, the one-dimensional eigenspace yields a column with constant modulus $1/\sqrt{n}$, so $n \|q\|_\infty^2 = 1$.

Therefore

$$(5) \quad M(Q^{(\text{ph})}) \leq \max \left\{ \mathbf{1}_{\{2|n\}}, \max_{1 \leq j \leq \lfloor (n-1)/2 \rfloor} 2 \cos^2(\Delta^*(n/\gcd(j, n))) \right\}.$$

Since $n' \mapsto \cos(\Delta^*(n'))$ is strictly increasing in n' in each congruence class of $n' \pmod{4}$, the inner maximum is achieved when n' is as large as possible, namely $n' = \max\{n/\gcd(j, n)\} = n$ (take any j coprime to n). Using the explicit values for $\Delta^*(n)$ from Lemma 7 gives

$$\max_j 2 \cos^2(\Delta^*(n/\gcd(j, n))) = \begin{cases} 2 \cos^2(\pi/n), & n \equiv 0 \pmod{4}, \\ 2 \cos^2(\pi/2n), & n \equiv 2 \pmod{4}, \\ 2 \cos^2(\pi/4n), & n \text{ odd}. \end{cases}$$

For $n \equiv 0 \pmod{4}$ we have $2 \cos^2(\pi/n) \geq 1$ with equality only at $n = 4$; for $n \equiv 2 \pmod{4}$ clearly $2 \cos^2(\pi/2n) > 1$; for odd n the $j = n/2$ column is absent. Thus the outer maximum in (5)

equals the displayed piecewise quantity, proving

$$M(Q^{(\text{ph})}) \leq \begin{cases} 2 \cos^2(\pi/n), & n \equiv 0 \pmod{4}, \\ 2 \cos^2(\pi/2n), & n \equiv 2 \pmod{4}, \\ 2 \cos^2(\pi/4n), & n \text{ odd}. \end{cases}$$

In order to demonstrate optimality (the lower bound), let us consider any orthogonal matrix Q obtainable from the cycle basis by independent rotations within each E_j (this is what we mean by “within the cycle family”). Fix j and write $n' = n/\gcd(j, n)$. By Lemma 7, *every* orthonormal pair drawn from E_j satisfies

$$\max\{n\|q\|_\infty^2 : q \in E_j, \|q\|_2 = 1\} \geq 2 \cos^2(\Delta^*(n')).$$

Indeed, fix j and recall that E_j is the two-dimensional eigenspace spanned by $\{u_j, v_j\}$. By Lemma 7, for *any* orthonormal pair (u, v) spanning E_j one has

$$\max\{\|u\|_\infty, \|v\|_\infty\} \geq \sqrt{\frac{2}{n}} \cos(\Delta^*(n')), \quad n' := \frac{n}{\gcd(j, n)}.$$

Squaring and multiplying by n yields

$$\max\{n\|u\|_\infty^2, n\|v\|_\infty^2\} \geq 2 \cos^2(\Delta^*(n')).$$

Since u and v are unit vectors in E_j , this shows the desired inequality. In other words, whichever orthonormal basis of E_j we pick, at least one vector in it has sup-norm squared of order $2 \cos^2(\Delta^*(n'))/n$. Thus the displayed inequality holds for *every* orthonormal pair drawn from E_j . (In the special case $j = n/2$ when n is even, $E_{n/2}$ is one-dimensional, and one can choose q with $|q(k)| \equiv 1/\sqrt{n}$, so $n\|q\|_\infty^2 = 1$; this agrees with $2 \cos^2(\Delta^*(2)) = 1$.)

Hence

$$\begin{aligned} M(Q) &\geq \max \left\{ \mathbf{1}_{\{2|n\}}, \max_j 2 \cos^2(\Delta^*(n/\gcd(j, n))) \right\} \\ &= \begin{cases} 2 \cos^2(\pi/n), & n \equiv 0 \pmod{4}, \\ 2 \cos^2(\pi/2n), & n \equiv 2 \pmod{4}, \\ 2 \cos^2(\pi/4n), & n \text{ odd}, \end{cases} \end{aligned}$$

where the equality follows by the same monotonicity and the presence (when $2 \mid n$) of the $j = n/2$ column with value $1 \leq$ the right-hand side. Combining the upper and lower bounds yields the asserted equality for $M(Q^{(\text{ph})})$ and proves optimality within the cycle family.

Finally, Lemma 10 with $M(Q^{(\text{ph})})$ as above gives the sufficient condition

$$1 + \sum_{j=2}^n \lambda_j > 1 - \frac{1}{M(Q^{(\text{ph})})},$$

which is precisely the stated $\delta_n^{(\text{ph})}$. □

Corollary 9 (Uniform phase bound). *As $\delta_* := \sup_{n \geq 3} \delta_n^{(\text{ph})} = \frac{1}{2}$, for every $n \geq 3$ and every Suleĭmanova list $(1, \lambda_2, \dots, \lambda_n)$ with $\lambda_j \leq 0$,*

$$1 + \sum_{j=2}^n \lambda_j > \delta_* (= \tfrac{1}{2}) \implies P = Q^{(\text{ph})} \Lambda (Q^{(\text{ph})})^\top,$$

P is symmetric, doubly stochastic, and entrywise non-negative.

5. A COHERENCE CRITERION

While the phase-optimised basis successfully removed the ‘ $8 \mid n$ ’ artefact, its coherence still approaches the maximum value of 2 as $n \rightarrow \infty$, preventing a uniform improvement on the $\delta < 1/2$ bound. To achieve a stronger result, we must look beyond the eigenbasis of the cycle graph. This section develops a general tool to guide this search. We first isolate the key property of the basis that governs the SDIEP bound, what we call its *coherence*, a measure of the peak magnitude of its entries, and prove that a lower coherence directly implies a better bound δ . Armed with this criterion, we then exhibit two families of ‘flat’ bases, constructed from Hadamard matrices, which possess the ideal coherence of $M(Q) = 1$. For dimensions where such matrices do exist, this allows us to prove the realisability of *all* Suleĭmanova spectra with a strictly positive trace sum.

Lemma 10 (Generic coherence criterion). *Let Q be any (real) orthogonal matrix with first column $q_0 = \mathbf{1}/\sqrt{n}$. Define*

$$M(Q) := \sup_{1 \leq j \leq n-1} \sup_{0 \leq k, l \leq n-1} n |q_j(k) q_j(l)| = n \max_{1 \leq j \leq n-1} \|q_j\|_\infty^2.$$

If $1 + \sum_{j=2}^n \lambda_j > 1 - \frac{1}{M(Q)}$ and $\lambda_j \leq 0$, then $P(\Lambda) = Q \Lambda Q^\top$ is symmetric, doubly stochastic, and entrywise non-negative.

Proof. As in Lemma 2, $P(\Lambda) \mathbf{1} = \mathbf{1}$ and

$$p_{kl} = \frac{1}{n} \left(1 + \sum_{j=1}^{n-1} \lambda_j n q_j(k) q_j(l) \right) \geq \frac{1}{n} \left(1 + M(Q) \sum_{j=1}^{n-1} \lambda_j \right)$$

since $\lambda_j \leq 0$ and $n q_j(k) q_j(l) \leq M(Q)$. □

The cycle basis is the standard Fourier basis for the cyclic group $\mathbb{Z}_n$. To achieve the minimal coherence $M = 1$ with a real orthogonal basis, we turn instead to the elementary abelian 2-group. General finite abelian 2-groups may have complex characters; the real $\{\pm 1\}$ character table used below is the one for the exponent-two group $G = (\mathbb{Z}/2\mathbb{Z})^d$. Its elements are binary vectors $\mathbf{x} = (x_1, \dots, x_d)$, and the group operation is vector addition modulo 2 (bitwise XOR).

The natural analogue of the Fourier basis on this group is the basis of its *real characters*. For each indexing vector $\mathbf{a} \in G$, there is a corresponding character $\chi_{\mathbf{a}} : G \rightarrow \{\pm 1\}$ defined by the dot product:

$$\chi_{\mathbf{a}}(\mathbf{x}) = (-1)^{\mathbf{a} \cdot \mathbf{x}} = (-1)^{\sum_{i=1}^d a_i x_i}.$$

The *Sylvester* (or *Walsh–Hadamard*) matrix H_n is constructed by indexing its rows and columns by the elements of G (typically in lexicographical order) and setting the entry at row $\mathbf{a}$ and column $\mathbf{x}$ to be $\chi_{\mathbf{a}}(\mathbf{x})$.

Remark 11 (Hadamard matrix orders). It is a classical open problem, known as the *Hadamard conjecture*, that Hadamard matrices exist for every order n equal to 1, 2, or a multiple of 4 [2]. Therefore, a Walsh–Hadamard matrix of order 3 does not exist.

A simple recursive construction, known as the Sylvester construction, generates these matrices for all orders $n = 2^d$:

$$H_1 = (1), \quad H_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H_{2n} = \begin{pmatrix} H_n & H_n \\ H_n & -H_n \end{pmatrix}.$$

For example, the matrix for $n = 4 = 2^2$ is constructed as follows:

$$H_4 = \begin{pmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{pmatrix} = \left(\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ \hline 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{array} \right).$$

By construction, the entries of H_n are all ± 1 and its columns are orthogonal. When normalised to $Q = H_n/\sqrt{n}$, we obtain an orthogonal matrix whose entries are all $\pm 1/\sqrt{n}$. This “flat” structure gives a coherence of $M(Q) = 1$, providing a powerful tool for the SDIEP as formalised in the following theorem.

Theorem 12 (Walsh–Hadamard bases from elementary abelian 2-groups). *If $n = 2^d$, let H_n be the Sylvester (Walsh–Hadamard) matrix of order n . After sign/permutation normalisation, $Q = H_n/\sqrt{n}$ is orthogonal with first column $\mathbf{1}/\sqrt{n}$ and entries $\pm 1/\sqrt{n}$, hence $M(Q) = 1$. Therefore, for every Suleĭmanova spectrum with $1 + \sum_{j=2}^n \lambda_j > 0$, the matrix $Q\Lambda Q^\top$ is symmetric and doubly stochastic.*

Theorem 13 (Hadamard orders via Paley-type constructions). *Whenever a (real) Hadamard matrix H of order n exists with first column $\mathbf{1}$, taking $Q = H/\sqrt{n}$ yields $M(Q) = 1$. Hence every Suleĭmanova spectrum with $1 + \sum_{j=2}^n \lambda_j > 0$ is realised by a symmetric doubly stochastic matrix via $Q\Lambda Q^\top$.*

Proof of Theorems 12 and 13 (combined). Let H be a real Hadamard matrix of order n . By multiplying columns by ± 1 and permuting them (operations preserving the Hadamard property), normalise so that the first column is $\mathbf{1}$. Set $Q := H/\sqrt{n}$. Then Q is real orthogonal, its first column is $q_0 = \mathbf{1}/\sqrt{n}$, and all entries satisfy $q_j(k) \in \{\pm 1/\sqrt{n}\}$.

Hence $M(Q) = \sup_{j \geq 1, k, l} n |q_j(k) q_j(l)| = 1$. By Lemma 10, whenever $1 + \sum_{j=2}^n \lambda_j > 1 - \frac{1}{M(Q)} = 0$ and $\lambda_j \leq 0$, the matrix $P(\Lambda) = Q\Lambda Q^\top$ is entrywise nonnegative. Symmetry is immediate, and

$$P(\Lambda) \mathbf{1} = Q\Lambda Q^\top \mathbf{1} = Q\Lambda(\sqrt{n} e_0) = \sqrt{n} Q e_0 = \mathbf{1}$$

so $P(\Lambda)$ is doubly stochastic. The Walsh–Hadamard case corresponds to $n = 2^d$; Paley-type constructions and other Hadamard orders give further instances. $\square$

Remark 14. If $M(Q) = 1$ (Hadamard orders) and $T := 1 + \sum_{j=2}^n \lambda_j > \varepsilon > 0$, then $p_{k\ell} \geq T/n$ analytically. In floating point arithmetic with unit roundoff u , a conservative acceptance test is $\min_{k,\ell} P_{k\ell} \geq -(cu)$ for a modest constant c depending on n and the BLAS implementation (we used $c = 10^{-12}$ in Section 8).

Remark 15. (i) The result requires the *strict* condition $1 + \sum \lambda_j > 0$. Our lower bound on the matrix entries is $p_{kl} \geq \frac{1}{n}(1 + \sum \lambda_j)$, which guarantees positivity under the strict inequality. The boundary case $1 + \sum \lambda_j = 0$ is not guaranteed to be realisable and is known to fail for certain spectra (e.g., $(1, 0, \dots, 0, -1)$ in odd dimensions).

(ii) Our use of Hadamard bases provides a constructive proof for *all* (not necessarily sorted) Suleřmanova lists in these dimensions, provided the trace sum is positive. This complements results such as that of Johnson and Paparella [4], which establish existence for *normalised* (sorted) Suleřmanova lists.

6. FURTHER REMARKS AND EXTENSIONS

Asymptotics (large n). For small angles, $\cos^2 \theta = 1 - \theta^2 + O(\theta^4)$, hence

$$1 - \frac{1}{2 \cos^2 \theta} = \frac{1}{2} - \frac{\theta^2}{2} + O(\theta^4).$$

For the canonical cycle basis, $\theta = \Delta_n = \frac{\pi}{4n} \rho(n)$ (Lemma 3), so

$$\delta_n = \begin{cases} \frac{1}{2}, & n \equiv 0 \pmod{8}, \\ \frac{1}{2} - \frac{\pi^2}{32n^2} \rho(n)^2 + O(n^{-4}), & n \not\equiv 0 \pmod{8}. \end{cases}$$

Explicitly: $\delta_n = \frac{1}{2} - \frac{\pi^2}{2n^2} + O(n^{-4})$ for $n \equiv 4 \pmod{8}$; $\delta_n = \frac{1}{2} - \frac{\pi^2}{8n^2} + O(n^{-4})$ for $n \equiv 2, 6 \pmod{8}$; and $\delta_n = \frac{1}{2} - \frac{\pi^2}{32n^2} + O(n^{-4})$ for odd n . For the phase-optimised basis (Theorem 8),

$$\delta_n^{(\text{ph})} = \begin{cases} \frac{1}{2} - \frac{\pi^2}{2n^2} + O(n^{-4}), & n \equiv 0 \pmod{4}, \\ \frac{1}{2} - \frac{\pi^2}{8n^2} + O(n^{-4}), & n \equiv 2 \pmod{4}, \\ \frac{1}{2} - \frac{\pi^2}{32n^2} + O(n^{-4}), & n \text{ odd}. \end{cases}$$

Tensor products and direct products of graphs. If $Q^{(1)}$ and $Q^{(2)}$ are orthogonal with first columns $\mathbf{1}/\sqrt{n_1}$ and $\mathbf{1}/\sqrt{n_2}$, then $Q := Q^{(1)} \otimes Q^{(2)}$ is orthogonal with first column $\mathbf{1}/\sqrt{n_1 n_2}$ and

$$M(Q) \leq M(Q^{(1)}) M(Q^{(2)}).$$

Thus Lemma 10 gives the sufficient sum-only condition

$$1 + \sum_{j=2}^{n_1 n_2} \lambda_j > 1 - \frac{1}{M(Q^{(1)}) M(Q^{(2)})}$$

for entrywise nonnegativity via $Q \Lambda Q^\top$. In particular, flat ($M = 1$) factors propagate.

7. LIMITATIONS AND OPEN PROBLEMS

For the canonical cycle basis, $M(Q) = 2$ exactly when $8 \mid n$ (Lemma 3), so no sum-only improvement below $\frac{1}{2}$ is possible there. Even when $8 \nmid n$, $M(Q) = 2 \cos^2 \Delta_n = 2(1 - O(n^{-2}))$, so the gain is $O(n^{-2})$ (Theorem 4). The phase-optimised basis yields $\delta_n^{(\text{ph})} < \frac{1}{2}$ for all n , but still not a single *uniform* $\delta < \frac{1}{2}$ independent of n .

Open problem. Is it possible to construct, for each n , a real orthogonal Q with $q_0 = \mathbf{1}/\sqrt{n}$ and $M(Q) \leq 2 - \varepsilon$ for some $\varepsilon > 0$ independent of n . By Lemma 10, this would give a universal $\delta < \frac{1}{2}$ for all (not necessarily normalised) Suleĭmanova lists.

8. NUMERICAL ANALYSIS

We complement the theoretical analysis with numerical experiments aimed at numerical linear algebra usage: construction of structured bases, certification of entrywise non-negativity, residual checks, timing of dense kernels, Monte-Carlo stress testing, and application of the resulting matrices as Markov-chain and consensus operators. All computations were performed in Python 3.12 with NumPy 1.26.4 [8]; the plotting used Matplotlib [9]. Complete code, data files, and exact reproduction instructions are available at <https://github.com/tomaszkania/sdiep-schur>, and the supplementary archive contains the scripts and CSV files used for the application tests below. We test entrywise non-negativity with the tolerance $\min_{k,\ell} P(\Lambda)_{k\ell} \geq -10^{-12}$, which is conservative relative to round-off.

8.1. Methodology. For each dimension n we construct orthogonal bases Q with Perron column $q_0 = \mathbf{1}/\sqrt{n}$ of four types: (i) canonical cycle (Section 3); (ii) phase-optimised cycle (Section 4); (iii) Walsh–Hadamard (powers of two); and (iv) Paley type I Hadamard (orders $q + 1$ with $q \equiv 3 \pmod{4}$ prime).

Random Suleĭmanova lists with prescribed trace sum $T = 1 + \sum_{j=2}^n \lambda_j$ are sampled as

$$(\lambda_2, \dots, \lambda_n) = -(1 - T)(X_2, \dots, X_n), \quad (X_2, \dots, X_n) \sim \text{Dirichlet}(\alpha \mathbf{1}),$$

with $\alpha = 1$ (uniform on the simplex). For each draw we compute $P(\Lambda) = Q\Lambda Q^\top$ and record success if $P(\Lambda) \geq 0$ entrywise. Monte-Carlo proportions are shown with Wilson [17] 95% confidence intervals; for $N = 1000$ trials the error bars are smaller than plotting symbols in our figures and are omitted for clarity.

8.2. Coherence verification. Figure 1 compares the empirical coherence $M(Q) = n \max_{j \geq 1} \|q_j\|_\infty^2$ to the theoretical values. The canonical cycle basis shows the mod-8 pattern with $M(Q) = 2$ exactly at $8 \mid n$; the phase-optimised basis removes this artefact and follows the piecewise form of Theorem 8. Walsh–Hadamard and Paley bases attain $M(Q) = 1$ as predicted by Lemma 10.

All discrepancies are at the level of floating-point round-off (below 5×10^{-14} in our experiments), reinforcing Lemmata 3 and 7.

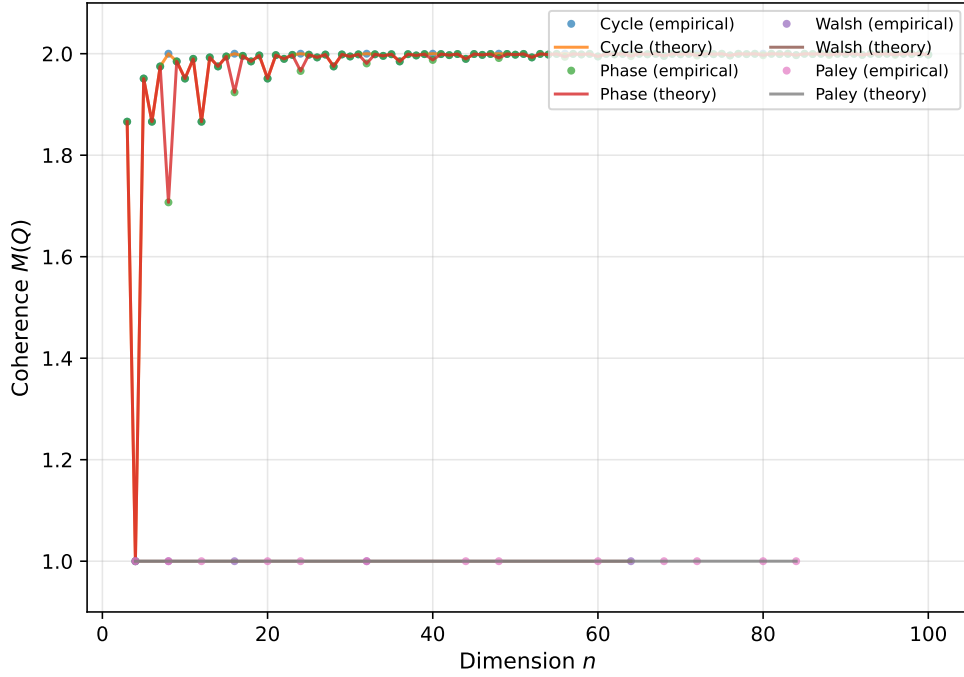


FIGURE 1. Empirical coherence $M(Q) = n \max_{j \geq 1} \|q_j\|_\infty^2$ versus dimension n for the four basis families implemented in the numerical code. The plot is a direct numerical verification of the closed-form coherence formulae proved in Sections 3–5: the canonical cycle basis attains the extremal value $M(Q) = 2$ exactly when $8 \mid n$; the phase-optimised cycle basis removes this artefact and yields $M(Q) < 2$ for every $n \geq 3$; and Walsh–Hadamard/Paley bases attain the optimal value $M(Q) = 1$. The numerical and theoretical values agree to floating-point round-off.

TABLE 1. Coherence verification: relative error statistics over the tested dimensions. For the cycle and phase-optimised families this covers $n = 3, \dots, 100$; for Walsh–Hadamard and Paley it covers the admissible orders in that range.

Basis	Count	Mean	Std	Max
Canonical cycle	98	1.46×10^{-15}	1.69×10^{-15}	8.22×10^{-15}
Phase-optimised	98	1.52×10^{-14}	1.16×10^{-14}	4.76×10^{-14}
Walsh–Hadamard	5	8.88×10^{-17}	1.22×10^{-16}	2.22×10^{-16}
Paley type I	13	1.37×10^{-16}	1.03×10^{-16}	2.22×10^{-16}

8.3. Success probability versus trace sum. Figure 2 plots the empirical success rate $\mathbb{P}[P(\Lambda) \geq 0]$ as a function of the trace sum $T = 1 + \sum_{j \geq 2} \lambda_j$ for representative sizes/bases. The panels are nearly uniformly near 1, i.e. random Dirichlet-uniform Suleĭmanova lists are realised with overwhelming

probability even *below* the sufficient thresholds δ_n (canonical) and $\delta_n^{(\text{ph})}$ (phase-optimised). This is expected: the bounds are *sufficient* (worst-case) and are attained by highly structured ‘spiky’ spectra (Proposition 5), whereas typical draws are far from these extremals. For Hadamard orders, success is 1 for all $T > 0$, in agreement with Theorems 12–13.

8.4. Application: reversible Markov chains with prescribed mixing rates. A symmetric doubly stochastic matrix is the transition matrix of a reversible Markov chain with uniform stationary distribution. Consequently, the Schur construction can be used as a numerical design method: prescribe the non-Perron eigenvalues, certify non-negativity, and obtain a Markov chain whose convergence is controlled by the second-largest eigenvalue modulus (SLEM); see, for instance, the standard spectral viewpoint in Markov-chain mixing theory [5].

We tested this design problem in the Hadamard order $n = 16$. For a fixed trace sum $T = 0.05$ and target rates $r \in \{0.25, 0.50, 0.75, 0.90\}$ we used

$$\lambda_2 = -r, \quad \lambda_3 = \dots = \lambda_{16} = -\frac{1 - T - r}{14}.$$

The Hadamard certificate gives $\min_{k,\ell} P_{k\ell} \geq T/n = 0.003125$ for every tested r . Figure 3 shows the total-variation distance from uniform for a point-mass initial distribution. The observed iteration counts increase with the prescribed SLEM, as expected from the spectral construction.

8.5. Application: average consensus with certified non-negative weights. The same matrices define linear average-consensus iterations

$$x^{(m+1)} = Px^{(m)},$$

where symmetry and double stochasticity preserve the arithmetic mean and entrywise non-negativity makes every update a convex averaging step. Such iterations are a standard numerical linear algebra model in distributed averaging and networked multi-agent systems [7, 6].

This test illustrates why the phase-optimised cycle basis is useful beyond a marginal theoretical improvement. Take $n = 24$, so $8 \mid n$ and the canonical cycle certificate has threshold $1/2$. We prescribe the one-spike spectrum

$$(1, -0.505, 0, \dots, 0), \quad T = 0.495.$$

Here T is below the canonical cycle threshold, but above the phase threshold $\delta_{24}^{(\text{ph})} \approx 0.491334$. The canonical cycle construction with this same spectrum has a negative entry $-4.17 \cdot 10^{-4}$, whereas the phase-optimised construction is certified and has minimum entry $3.00 \cdot 10^{-4}$. Figure 4 shows the resulting consensus error for a random initial vector.

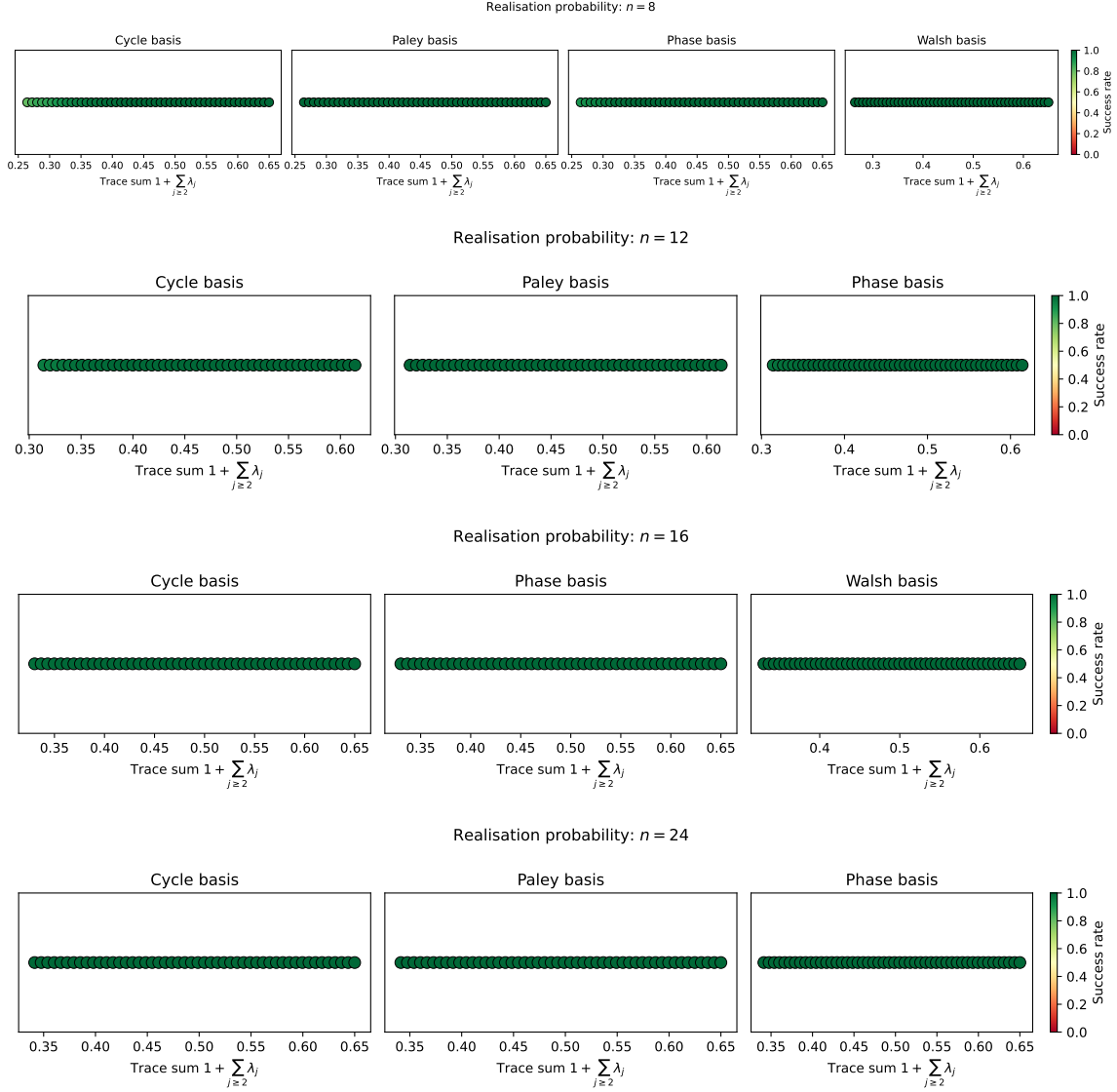


FIGURE 2. Empirical realisation probability $\mathbb{P}[P(\Lambda) \geq 0]$ as a function of the trace sum $T = 1 + \sum_{j \geq 2} \lambda_j$ for selected dimensions. Each block corresponds to one dimension ($n = 8, 12, 16, 24$), and within each block we compare the basis families implemented in the repository <https://github.com/tomaszkania/sdiep-schur>. The main numerical message is a typical-case/worst-case distinction: the analytic quantities δ_n and $\delta_n^{(\text{ph})}$ are worst-case sufficient thresholds, whereas for Dirichlet-sampled Suleĭmanova lists the Schur template succeeds with high probability already below those values. In Hadamard orders (visible here at $n = 16$), the success rate is numerically 1 for all tested $T > 0$, in agreement with Lemma 10.

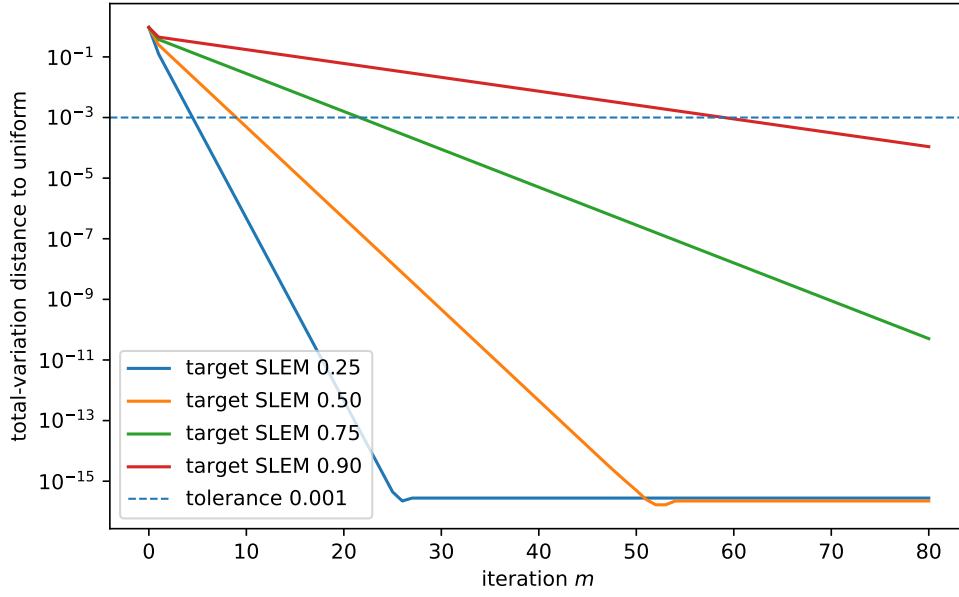


FIGURE 3. Reversible Markov-chain design in the Hadamard order $n = 16$. Each curve uses a spectrum with trace sum $T = 0.05$ and prescribed SLEM r . The constructed transition matrices are symmetric, doubly stochastic, and entrywise non-negative by the $M(Q) = 1$ certificate. The total-variation distance from the uniform stationary distribution decays at a rate governed by the chosen non-Perron spectrum.

TABLE 2. Application diagnostics. The Markov rows use the Hadamard basis at $n = 16$ and tolerance 10^{-3} for total variation. The consensus rows use $n = 24$ and tolerance 10^{-6} for the relative disagreement norm; the canonical row is included as a failed non-negativity comparator for the same spectrum. Row-sum and symmetry residuals in the successful rows are at round-off level.

Case	n	T	SLEM	$\min P_{k\ell}$	spectral residual	iter.
Hadamard, $r = 0.25$	16	0.050	0.250	$3.125 \cdot 10^{-3}$	$4.72 \cdot 10^{-16}$	5
Hadamard, $r = 0.50$	16	0.050	0.500	$3.125 \cdot 10^{-3}$	$5.55 \cdot 10^{-16}$	9
Hadamard, $r = 0.75$	16	0.050	0.750	$3.125 \cdot 10^{-3}$	$2.22 \cdot 10^{-16}$	22
Hadamard, $r = 0.90$	16	0.050	0.900	$3.125 \cdot 10^{-3}$	$2.22 \cdot 10^{-16}$	59
Phase cycle	24	0.495	0.505	$3.00 \cdot 10^{-4}$	$2.67 \cdot 10^{-16}$	17
Canonical cycle	24	0.495	0.505	$-4.17 \cdot 10^{-4}$	$2.22 \cdot 10^{-16}$	–

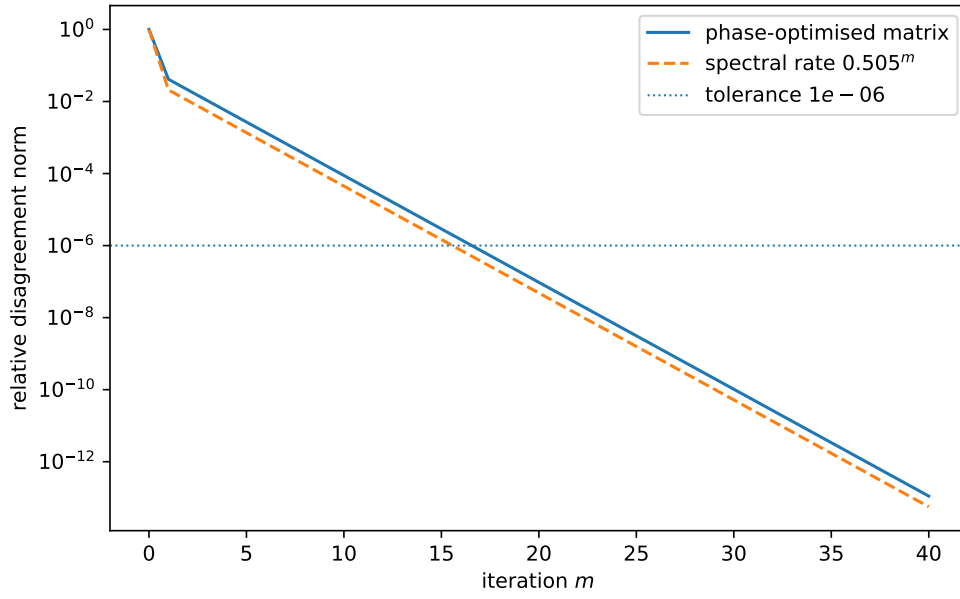


FIGURE 4. Average-consensus test for the phase-optimised cycle basis at $n = 24$ with spectrum $(1, -0.505, 0, \dots, 0)$ and trace sum $T = 0.495$. The phase basis gives a valid non-negative averaging matrix even though the same trace sum lies below the canonical cycle threshold. The relative disagreement norm decays according to the prescribed SLEM 0.505.

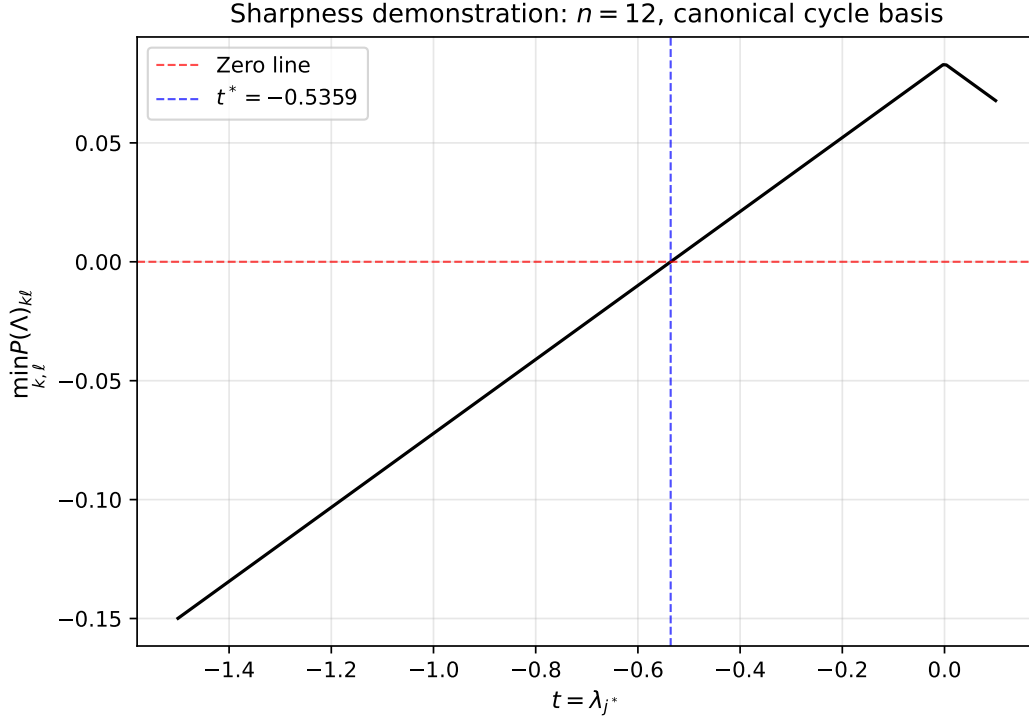


FIGURE 5. Sharpness test for the canonical cycle basis at $n = 12$ using one-spike spectra: $\lambda_{j^*} = t < 0$ with $j^* = 1$, and $\lambda_j = 0$ for $j \notin \{0, j^*\}$. For each t we evaluate the minimum entry $\min_{k,\ell} P(\Lambda)_{k\ell}$. Proposition 5 predicts an affine dependence on t and a zero crossing at $t^* = -1/(2 \cos^2(\Delta_{12})) = -1/(2 \cos^2(\pi/12)) \approx -0.5359$. The numerical curve crosses 0 at the same value (up to grid resolution and floating-point tolerance), confirming that the trace-sum bound for the canonical cycle basis is genuinely sharp within this template.

8.6. Sharpness at the canonical cycle threshold. To exhibit the worst-case behaviour predicted by Proposition 5, we consider one-spike spectra in $n = 12$: set $\lambda_{j^*} = t < 0$ with $j^* = 1$ (coprime to n) and $\lambda_j = 0$ for $j \notin \{0, j^*\}$. By Lemma 3, since $12 \equiv 4 \pmod{8}$ we have $\rho(12) = 4$ and thus $\Delta_{12} = \pi\rho(12)/(4 \cdot 12) = \pi/12$. Proposition 5 then predicts that $p_{k^*k^*}$ crosses zero at

$$t^* = -\frac{1}{2 \cos^2(\Delta_{12})} = -\frac{1}{2 \cos^2(\pi/12)} \approx -0.535898 \dots$$

implying $1 + \sum_{j=2}^n \lambda_j = 1 + t^* \approx 0.464102$, in agreement with δ_{12} from Theorem 4.

Figure 5 shows that $\min_{k,\ell} P(\Lambda)_{k\ell}$ varies linearly in t and crosses 0 at $t \approx -0.536$, confirming the sharpness of Theorem 4 within the cycle template.

8.7. Computational cost. Constructing the canonical or phase-optimised cycle basis explicitly as a dense matrix costs $O(n^2)$ floating-point operations, dominated by evaluating trigonometric functions on an $n \times n$ grid. In contrast, Walsh–Hadamard bases can be generated recursively with $O(n^2)$ memory operations and admit fast matrix–vector multiplies via the fast Walsh–Hadamard transform.

Evaluating $P(\Lambda) = Q\Lambda Q^\top$ as a dense matrix uses Level-3 BLAS operations: forming $Q\Lambda$ costs $O(n^2)$ flops (diagonal scaling), and the product $(Q\Lambda)Q^\top$ costs $O(n^3)$ flops using standard dense matrix multiplication. For moderate sizes (e.g. $n \leq 128$) this is inexpensive on a laptop CPU, and for repeated evaluations the structured bases (FFT/FWHT) allow significantly faster matrix–vector products without forming $P(\Lambda)$ explicitly. Figure 6 summarises the measured construction and dense evaluation times in the tested range.

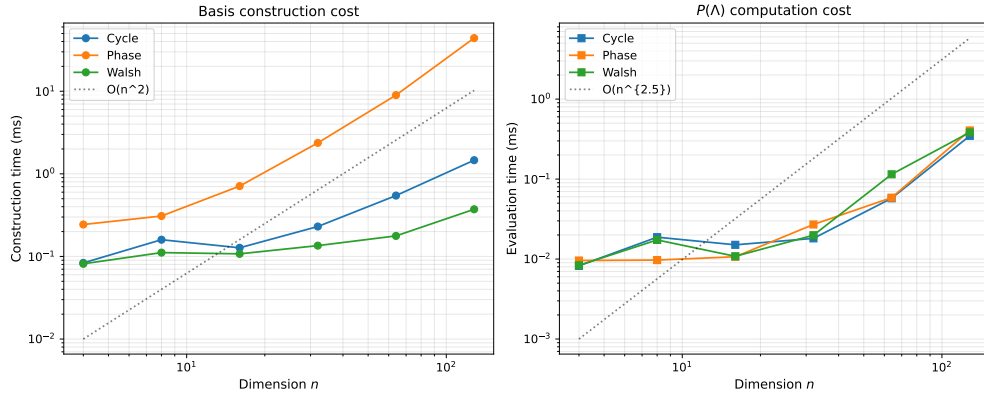


FIGURE 6. Measured construction and dense Schur-evaluation times for the implemented basis families. The timing experiment confirms the expected dense behaviour: basis construction is quadratic in storage, while forming the full matrix $P = Q\Lambda Q^\top$ is dominated by a dense matrix–matrix product. Walsh–Hadamard bases are especially cheap to construct recursively and can also be applied matrix-free by the fast Walsh–Hadamard transform.

Remark 16 (Fast application without forming $P(\Lambda)$). For many numerical tasks one does not need the full matrix $P(\Lambda)$ explicitly, but only products $P(\Lambda)x$. These can be computed as

$$P(\Lambda)x = Q(\Lambda(Q^\top x)).$$

Thus the cost is dominated by applying Q and $Q^\top$. For Walsh–Hadamard bases this can be done by the fast Walsh–Hadamard transform in $O(n \log n)$ operations, and for Fourier/cycle-type bases by FFT-based routines after passing to the corresponding complex representation. In this sense the proposed constructions yield not only explicit realisations but also structured linear operators suitable for large-scale iterative computations.

8.8. Discussion. The experiments support the analytic findings of the paper. Indeed,

- The empirical values of $M(Q)$ match the formulae exactly: $M(Q_{\text{cycle}})$ attains 2 precisely at $8 \mid n$; phase optimisation removes this artefact and lowers $M(Q)$ across all n ; Hadamard families achieve the optimal $M(Q) = 1$.
- For Dirichlet-uniform random Suleĭmanova lists with a given trace sum, the Schur template succeeds with overwhelming probability even *below* the sufficient thresholds δ_n and $\delta_n^{(\text{ph})}$. This is expected: the bounds are worst-case and are saturated by highly structured ‘spiky’ spectra, not by typical random draws.
- In the one-spike family, the minimum entry varies *linearly* in the spike height t and crosses zero at $t^* = -1/(2 \cos^2 \Delta_n)$, exactly as predicted. Hence the cycle-basis threshold cannot be improved by a sum-only condition within that template.
- When a Hadamard order is available, choosing Q from a Hadamard family (Walsh or Paley) guarantees realisation for every strict Suleĭmanova list ($M(Q) = 1$, hence $\delta = 0$). Otherwise, the phase-optimised cycle basis improves uniformly over the canonical choice and removes the $8 \mid n$ barrier.
- The Markov-chain and consensus tests show how the constructions can be used as numerical design tools: one prescribes spectral behaviour, receives a certified stochastic matrix, and verifies the row-sum, symmetry, spectral, and non-negativity residuals.

All code and data to reproduce Figures 1–6, including the application experiments, are available at <https://github.com/tomaszkania/sdiep-schur>. The supplementary archive contains the CSV files and the script `nla_application_experiments.py` used for Figures 3 and 4.

APPENDIX: VALUES AND VISUALISATION OF THE PHASE-OPTIMISED BOUND

Using Theorem 8, concrete values (rounded) are:

n	$\delta_n^{(\text{ph})}$	n	$\delta_n^{(\text{ph})}$	n	$\delta_n^{(\text{ph})}$	n	$\delta_n^{(\text{ph})}$
3	0.4641016151	4	0	5	0.4874571845	6	0.4641016151
7	0.4936524043	8	0.4142135624	9	0.4961728669	10	0.4874571845
12	0.4641016151	16	0.4802169351	20	0.4874571845	24	0.4913338099
30	0.4986267123	40	0.4969030207	60	0.4986267123	100	0.4995061949

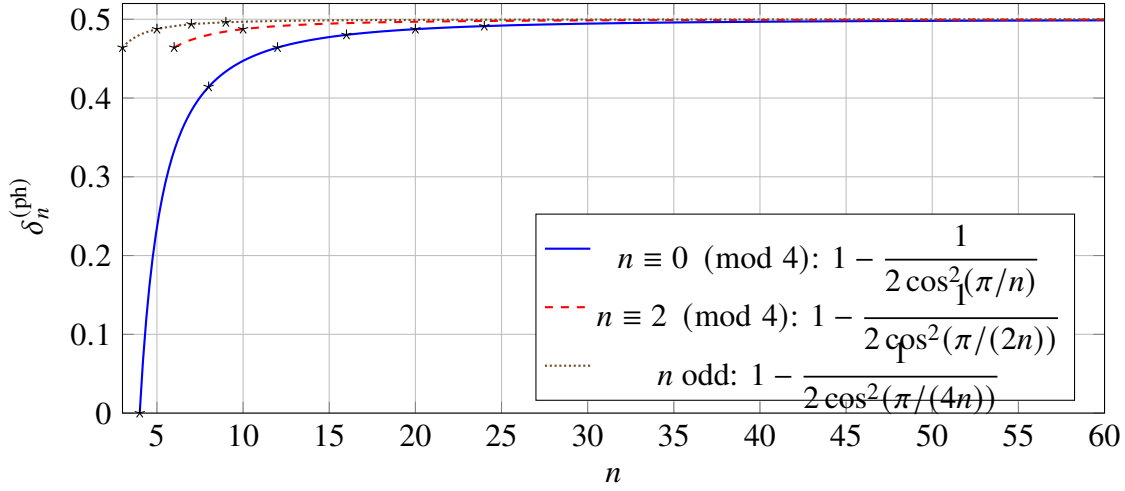


FIGURE 7. Phase-optimised threshold $\delta_n^{(\text{ph})}$ by congruence class. Curves are envelopes; only integer n on the labelled classes apply.

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